Resolved referee R043 [Gap; revised] – The annulus equality needs a proof.
Original source line 1132. Specify that A and B each have one VF-coordinate. The finite limit is a singleton $\{b\}$ by [17, Lemma 9.4]; nonemptiness follows because the values are nonempty on a punctured neighborhood. The equality of whole thin annuli needs more explanation. Here is a repair. The valuation calculation in the proof of [17, Lemma 8.3], applied to $x \mapsto \mathrm{val}(f(x) - b)$, gives

$$\mathrm{val}(f(x) - b) = q\,\mathrm{val}(x - a) + c \qquad (q \in \mathbb{Q}_{>0})$$

on a sufficiently small punctured neighborhood: each unary polynomial has eventually monomial valuation, and the limit forces positive slope. Remove the at most one exceptional input with $f(x) = b$. By C-minimality, B contains a punctured neighborhood of b. The inverse has a unique projective limit there, necessarily a, since points $x \longrightarrow a$ give $f(x) \longrightarrow b$. Apply the same valuation calculation to the inverse. The resulting two affine maps of tail radii are inverse, so every sufficiently small target shell consists exactly of images of its corresponding source shell. Taking their unions proves the asserted punctured-disc equality. Thus the lemma is repairable; an additional assertion that $q = 1$ is unnecessary.

Author resolution. The arity objection is withdrawn: the explicit subsets $A, B \subseteq \mathrm{VF}$ are already unary, and no amendment is needed. The author agrees that the original proof is quick and that the remaining comments spell out details. The expanded proof makes the singleton limit and the inverse tail-radius laws explicit, giving equality of whole thin annuli. The statement and proof strategy are retained; this is an expository expansion, not a correction of a false claim.

Definition 2.52. Let $p : A \longrightarrow \Gamma$ be a definable function. We say that p is an *$\mathfrak{o}$-partition* of A if, for every $a \in A$, the function p is constant on $\mathfrak{o}(a, p(a)) \cap A$.

This makes sense even when A has no VF-coordinates, in which case any definable function $A \longrightarrow \Gamma$ is an $\mathfrak{o}$-partition of A.

Lemma 2.53. *Let p be an $\mathfrak{o}$-partition of A. Suppose that A_{RV} is doubly bounded and A_t is closed and bounded for every $t \in A_{\mathrm{RV}}$. Then $p(A)$ is doubly bounded.*

Proof. First suppose that A has no RV-coordinates. The empty set is immediate. Choose $\eta \in \Gamma$ such that $A \subseteq \mathcal{O}_\eta^n$. If $p(a) < \eta$ for some $a \in A$, then $A \subseteq \mathfrak{o}(a, p(a))$, so the partition property makes p constant on A. Otherwise $p(A) \subseteq [\eta, \infty)$. In either case $p(A)$ is bounded below.

Suppose for contradiction that $p(A)$ is unbounded above. We may temporarily enlarge the parameter base to a small VF-generated base, as in Lemma 2.48; the desired bound is a geometric conclusion. For every $c \in \mathcal{M} \setminus \{0\}$ the definable set

$$A_{\mathrm{val}(c)} = \{x \in A : p(x) > \mathrm{val}(c)\}$$

is nonempty. Elementarity of the algebraic-closure model over this base together with c, followed by compactness, supplies a definable finitary function $F : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{P}(A)$ with finite nonempty values $F(c) \subseteq A_{\mathrm{val}(c)}$. Continuity of this choice is not needed.

By Lemma 2.45, F has a finite minimal limit set $L \subseteq A$, nonempty because its values are nonempty arbitrarily close to 0. Choose $e \in L$. Minimality implies that, for arbitrarily small nonzero c, $F(c)$ meets $\mathfrak{o}(e, p(e))$. Choose such a c with $\mathrm{val}(c) > p(e)$ and $x \in F(c) \cap \mathfrak{o}(e, p(e))$. The partition property at e gives $p(x) = p(e)$, whereas the choice gives $p(x) > \mathrm{val}(c) > p(e)$, a contradiction. This proves the upper bound.

For the general case, each nonempty fiber image $p(A_t)$ is bounded above and below by the case just proved, with t temporarily named. Definable completeness in Γ gives finite endpoints

$$\alpha_t = \inf p(A_t), \qquad \beta_t = \sup p(A_t).$$